Data, center, and spread

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Write $\frac{1}{2}$ as a decimal and a percent.

2. Write $\frac{1}{4}$ as a decimal and a percent.

3. Write $\frac{3}{4}$ as a decimal and a percent.

4. Write $\frac{1}{8}$ as a decimal and a percent.

5. Find $|-12|$ and describe it as a distance.

6. Find $2.1 \div 0.07$.

7. $72 \div 6$

8. Find the MAD of 3, 3, 9, 9.

9. Find the range of 2, 5, 8, 11.

10. Find Q1, Q3, and IQR for 1, 2, 4, 5, 7, 8, 10, 11. Exclude the median if the count is odd.

11. Find the MAD of 1, 3, 5, 7.

12. A: 3, 4, 5. B: 1, 4, 7. Compare their means and ranges.

13. Zoe averages −3, −1, 1, and 3 to find MAD. She gets 0. What should she average?

14. Choose the statistical question: (A) How many pages are in one named book? (B) How many pages are in each of 11 books? Explain.

15. Make a dot plot for 4, 4, 5, 6, 6, 6, 8. State the total dots and most frequent value.

16. Find histogram frequencies for 3, 4, 5, 5, 6, 7, 8, 8 using bins 3 to under 5, 5 to under 7, and 7 to under 9.

17. Find the mean of 5, 7, 9, 11. Explain it as a fair share.

18. Order 11, 5, 8, 7, 8, 12 and find the median.

CHALLENGE / Explain your plan and check the result

C1. Data are 1, 2, 4, 5, 7, 8, 10, 11. Find the five-number summary and IQR. Replace 11 with 31 and recompute only the range and IQR. Explain why one spread measure changes and the other does not.

C2. Two box plots have summaries A: (0, 2, 5, 8, 10) and B: (0, 4, 5, 6, 10). Compare their medians, ranges, and IQRs. Can you determine which has more observations from these summaries alone? Explain.